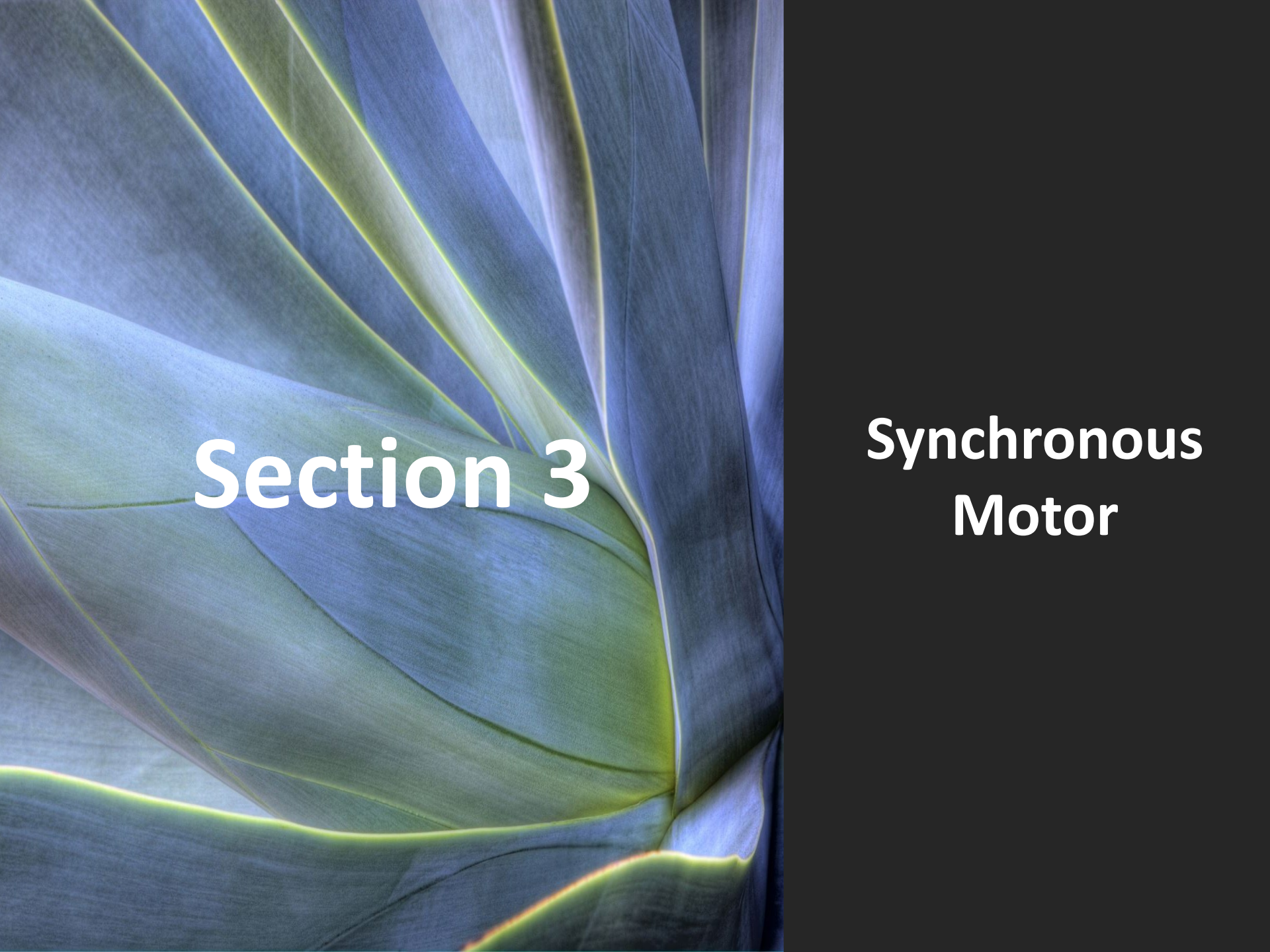


The background of the slide is an abstract composition of flowing, organic shapes in shades of blue, teal, and green. These shapes overlap and curve, creating a sense of movement and depth. The colors are vibrant and saturated, with some areas appearing lighter and others darker, suggesting a light source from the side.

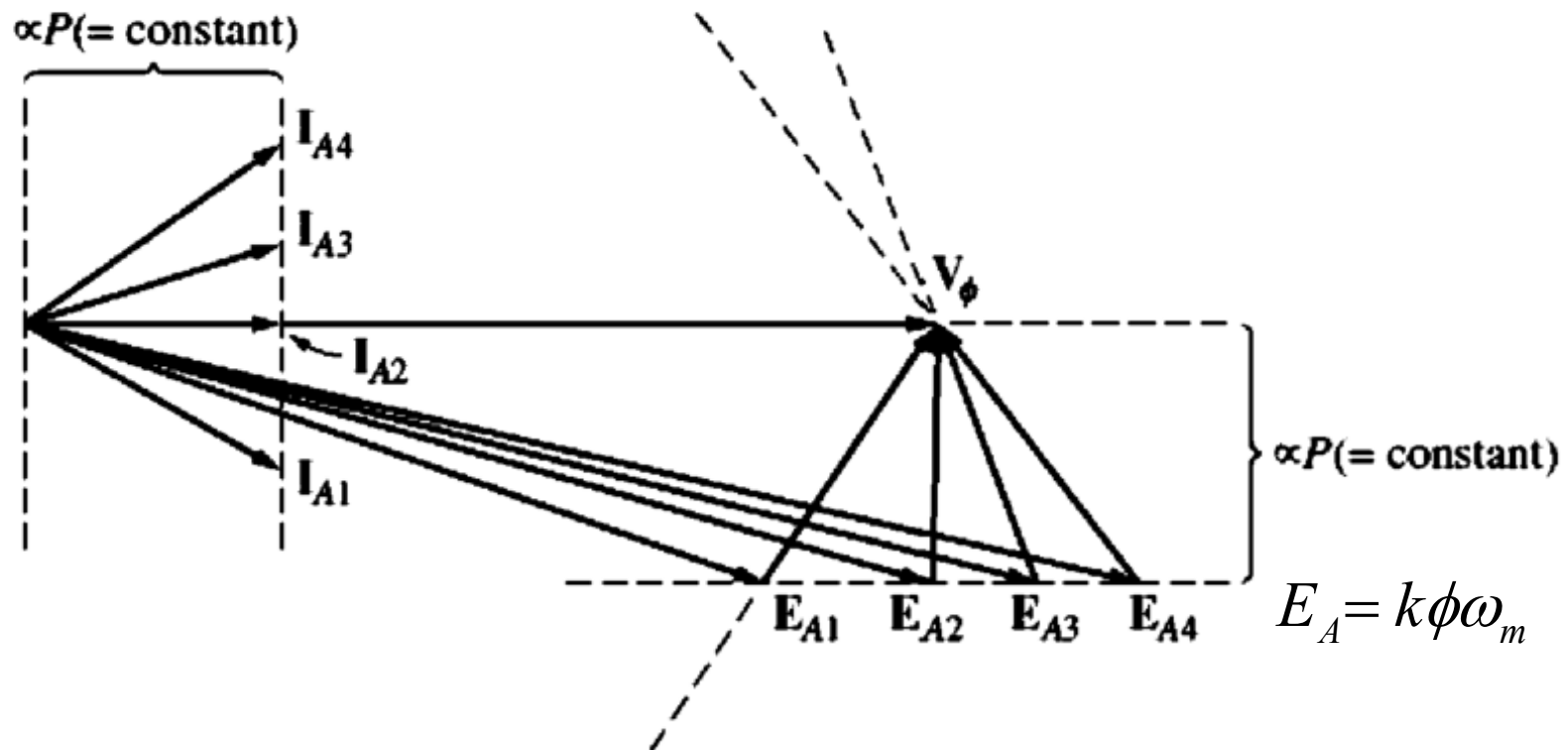
Section 3

Synchronous Motor

12. The Effect of Field Current Changes

What effect does a change in field current have on a synchronous motor?

- The field current controls the magnitude of the internal generated voltage (E_A).
- It does not affect the real power supplied by the motor. Power supplied by the motor changes only when the shaft load torque changes.
- When the field current is increased, E_A must increase, but it can only do so by sliding out along the line of constant power.



12. The Effect of Field Current Changes

1. Under-excited region (left side)

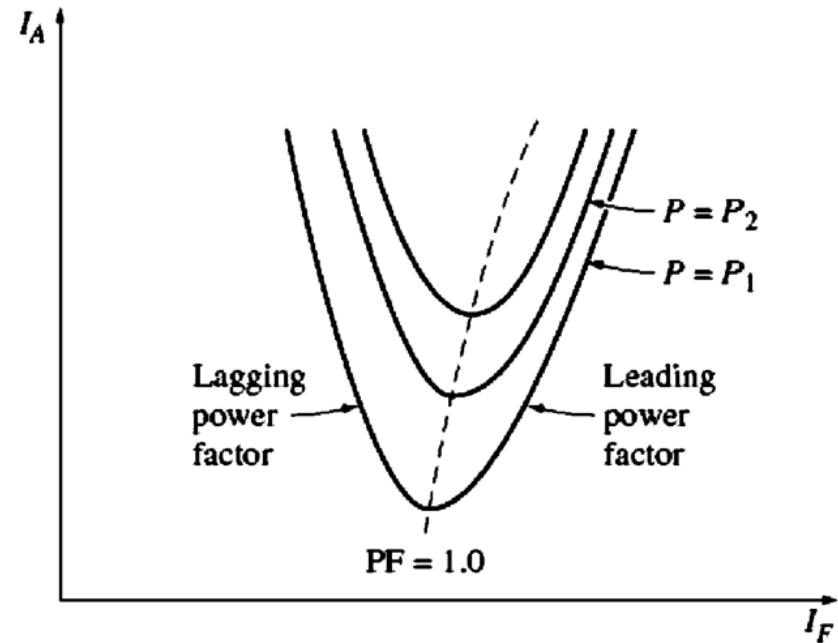
- The motor draws more armature current.
- Power factor is lagging.
- It needs extra magnetizing current from the supply.

2. Unity power factor (PF = 1)

- Exact field current required for unity power factor
- Armature current is minimum
- Motor neither leads nor lags
- This is the most efficient operating point

3. Over-excited region (right side)

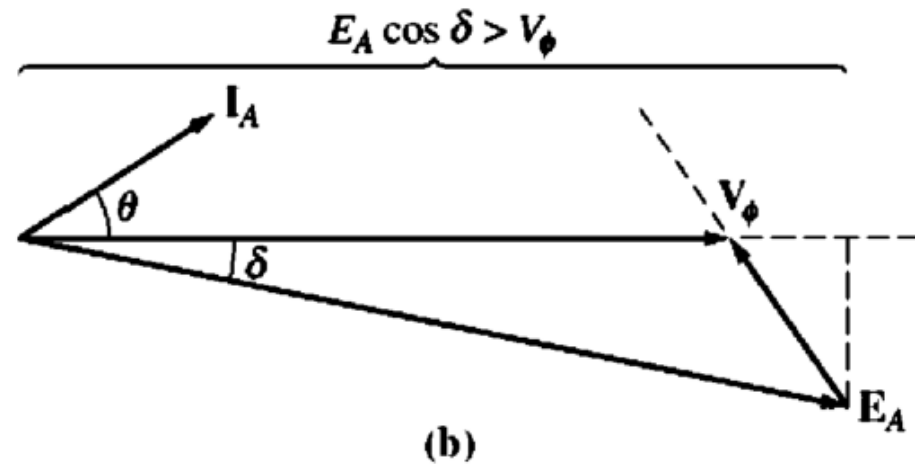
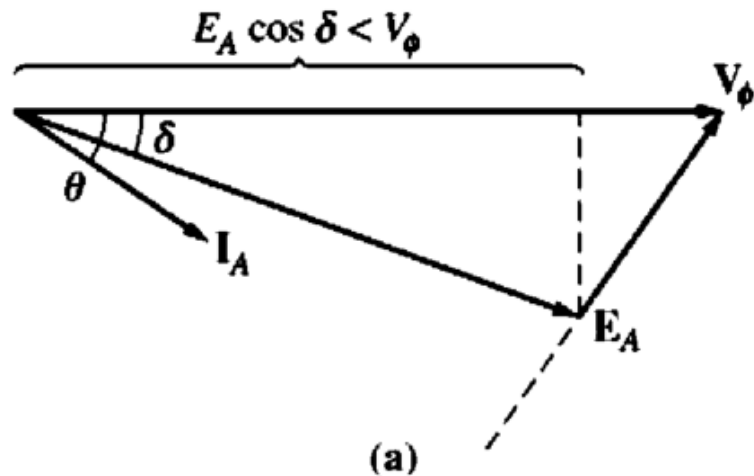
- Field current too high
- Power factor becomes leading
- Motor supplies reactive power (behaves like a capacitor)



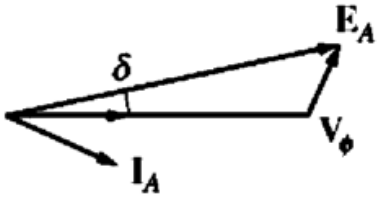
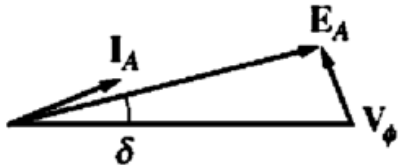
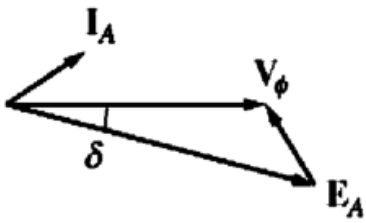
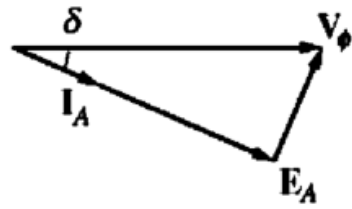
Synchronous motor V curve

12. The Effect of Field Current Changes

- When the projection of the phasor E_A on V_ϕ , $E_A \cos \delta$ is shorter than V_ϕ itself, a synchronous motor has a lagging current and consumes Q . Since the field current is small in this situation, the motor is said to be underexcited.
- On the other hand, when the projection of E_A on V_ϕ is longer than V_ϕ itself, a synchronous motor has a leading current and supplies Q to the power system. Since the field current is large in this situation, the motor is said to be overexcited.

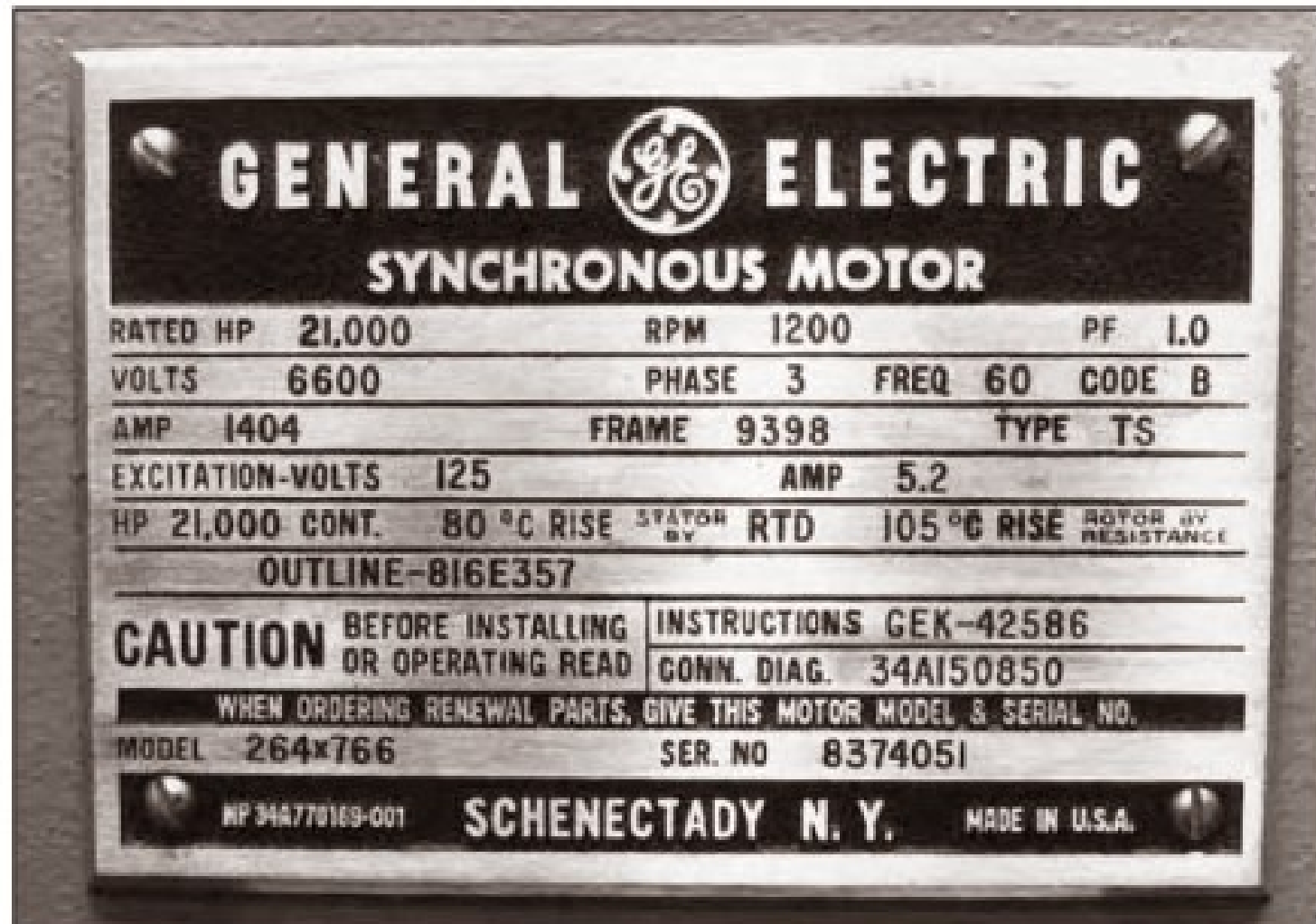


13. Synchronous Generator and Synchronous Motor

	Supply reactive power Q $E_A \cos \delta > V_\phi$	Consume reactive power Q $E_A \cos \delta < V_\phi$
Supply power P Generator E_A leads V_ϕ		
Consume power P Motor E_A lags V_ϕ		

14. Synchronous Motor Rating

Since synchronous motors are the same physical machines as synchronous generators, the basic machine ratings are the same.



Example 3

The 208-V, Δ -connected, 60-Hz synchronous motor is supplying a 15-hp load with an initial power factor of 0.85 PF lagging. It has a synchronous reactance of 2.5Ω and a negligible armature resistance. Its friction and windage losses are 1.5 kW, and its core losses are 1.0 kW.

1. Sketch the initial phasor diagram of this motor and find the values I_A and E_A ?
2. If the motor's flux is increased by 25 percent, sketch the new phasor diagram of the motor. What are E_A , I_A and the power factor of the motor now?

Example 4

The infinite bus in the following Figure, it operates at 480 V. Load 1 is an induction motor consuming 100 kW at 0.78 PF lagging, and load 2 is an induction motor consuming 200 kW at 0.8 PF lagging. Load 3 is a synchronous motor whose real power consumption is 150 kW.

1. If the synchronous motor is adjusted to operate at 0.85 PF lagging, what is the transmission line current in this system?
2. If the synchronous motor is adjusted to operate at 0.85 PF leading, what is the transmission line current in this system?
3. Assume that the transmission line losses are given by:
 $P_{LL} = 3I_L^2 R_L$ line loss, where LL stands for line losses.
Compare the transmission losses in the two cases?

